

Problem 1: First Customer Purchase

An e-commerce website has a **20% probability** that any visitor makes a purchase.

What is the probability that the **first purchase occurs on the 4th visitor**?

Solution

Given:

$$p = 0.20$$

Therefore:

$$q = 1 - p = 0.80$$

We want the first success on the 4th trial.

That means:

- Visitor 1 → No purchase
- Visitor 2 → No purchase
- Visitor 3 → No purchase
- Visitor 4 → Purchase

Therefore:

$$\begin{aligned}P(X = 4) &= q^{4-1}p \\P(X = 4) &= (0.8)^3(0.2) \\P(X = 4) &= 0.512(0.2)\end{aligned}$$

$$P(X = 4) = 0.1024$$

Answer

$$10.24\%$$

Data Science Interpretation

There is a **10.24% probability** that the first purchase will occur exactly with the 4th visitor.

This can be useful for analyzing **customer conversion behavior**.

Problem 2: First Advertisement Click

A website has a probability of **10%** that a visitor clicks an advertisement.

Find the probability that the **first click occurs on the 5th visitor**.

Solution

Given:

$$\begin{aligned}p &= 0.10 \\q &= 0.90\end{aligned}$$

Using:

$$P(X = x) = q^{x-1}p$$

For $x = 5$:

$$\begin{aligned}P(X = 5) &= (0.9)^4(0.1) \\P(X = 5) &= 0.6561(0.1)\end{aligned}$$

$$P(X = 5) = 0.06561$$

Answer

$$6.561\%$$

So there is approximately a **6.56% probability that the first advertisement click occurs on the 5th visitor**.

Problem 3: First Successful Recommendation

A recommendation system successfully recommends a product to a user with probability **0.25**.

Find the probability that the **first successful recommendation occurs on the 3rd attempt**.

Solution

Given:

$$\begin{aligned}p &= 0.25 \\q &= 0.75\end{aligned}$$

We need:

$$P(X = 3)$$

Therefore:

$$\begin{aligned} P(X = 3) &= q^{3-1}p \\ P(X = 3) &= (0.75)^2(0.25) \\ P(X = 3) &= 0.5625(0.25) \end{aligned}$$

$$P(X = 3) = 0.140625$$

Answer

$$14.06\%$$

Interpretation

There is approximately a **14.06% probability that the first successful recommendation occurs on the third attempt.**

Problem 4: First Fraudulent Transaction

Suppose a banking system estimates that the probability of a transaction being fraudulent is 2%.

Find the probability that the **first fraudulent transaction occurs on the 10th transaction.**

Solution

Given:

$$\begin{aligned} p &= 0.02 \\ q &= 0.98 \end{aligned}$$

We need:

$$P(X = 10)$$

Using:

$$P(X = x) = q^{x-1}p$$

Therefore:

$$\begin{aligned} P(X = 10) &= (0.98)^9(0.02) \\ &\approx 0.8337(0.02) \end{aligned}$$

$$P(X = 10) \approx 0.01667$$

Answer

$$1.67\%$$

Data Science Interpretation

There is approximately a **1.67% probability** that the first fraudulent transaction will occur exactly on transaction number 10.

Problem 5: Expected Number of Customers Until First Purchase

An online store has a conversion probability of:

$$p = 0.20$$

Find the **expected number of visitors until the first purchase**.

Solution

For a geometric distribution:

$$E(X) = \frac{1}{p}$$

Therefore:

$$E(X) = \frac{1}{0.20}$$

$$E(X) = 5$$

Answer

The expected number of visitors until the first purchase is:

$$5$$

Interpretation

On average, the company can expect the **first purchase after about 5 visitors**.

This does **not** mean that every fifth visitor will purchase. It is a long-run average.

Problem 6: Expected Number of Predictions Until First Error

A machine-learning classification model has an error probability of 5%.

How many predictions are expected before the first error occurs?

Solution

Given:

$$p = 0.05$$

For a geometric distribution:

$$E(X) = \frac{1}{p}$$

Therefore:

$$E(X) = \frac{1}{0.05}$$

$$E(X) = 20$$

Answer

The expected number of predictions until the first error is:

$$20$$

Data Science Interpretation

If the error probability remains 5%, the first error occurs after **20 predictions on average**.

Again, this is an expected value, not a guarantee.

Problem 7: Probability of First Churn on the 6th Customer

A customer-churn model estimates that a randomly selected customer has a **15% probability of churning** within a specified period.

Find the probability that the **first churn occurs at customer number 6**.

Solution

Given:

$$\begin{aligned}p &= 0.15 \\q &= 0.85\end{aligned}$$

We need:

$$P(X = 6)$$

Therefore:

$$\begin{aligned}P(X = 6) &= (0.85)^5(0.15) \\&\approx 0.4437(0.15)\end{aligned}$$

$$P(X = 6) \approx 0.06655$$

Answer

$$6.66\%$$

Problem 8: Probability of First Spam Email

Suppose the probability that an incoming email is spam is:

$$p = 0.10$$

Find the probability that the **first spam email occurs on or before the 5th email**.

Understanding "On or Before"

We need:

$$P(X \leq 5)$$

Instead of calculating five probabilities separately, use the complement:

$$P(X \leq 5) = 1 - P(X > 5)$$

For a geometric distribution:

$$P(X > k) = (1 - p)^k$$

Therefore:

$$P(X > 5) = (0.9)^5$$

$$P(X > 5) = 0.59049$$

Hence:

$$P(X \leq 5) = 1 - 0.59049$$

$$P(X \leq 5) = 0.40951$$

Answer

40.95%

Interpretation

There is approximately a **40.95% probability that at least one spam email will appear within the first 5 emails.**

Problem 9: At Least One Fraud Before the 20th Transaction

The probability of fraud for each transaction is:

$$p = 0.02$$

Find the probability of observing **at least one fraudulent transaction in the first 20 transactions.**

Solution

We want:

$$P(X \leq 20)$$

Use the complement:

$$P(X \leq 20) = 1 - P(X > 20)$$

For a geometric distribution:

$$P(X > 20) = (1 - p)^{20}$$

Therefore:

$$P(X \leq 20) = 1 - (0.98)^{20}$$
$$(0.98)^{20} \approx 0.6676$$

Thus:

$$P(X \leq 20) = 1 - 0.6676$$

$$P(X \leq 20) \approx 0.3324$$

Answer

33.24%

Data Science Interpretation

There is approximately a **33.24% probability of seeing at least one fraudulent transaction within the first 20 transactions.**

Problem 10: Variance of Waiting Time

A website visitor has a probability of **25%** of clicking an advertisement.

Find the:

1. Expected number of visitors until the first click
2. Variance
3. Standard deviation

Solution

Given:

$$p = 0.25$$

Mean

$$E(X) = \frac{1}{p}$$
$$E(X) = \frac{1}{0.25}$$

$$E(X) = 4$$

Variance

$$\begin{aligned} \text{Var}(X) &= \frac{1-p}{p^2} \\ \text{Var}(X) &= \frac{1-0.25}{(0.25)^2} \\ \text{Var}(X) &= \frac{0.75}{0.0625} \end{aligned}$$

$$\text{Var}(X) = 12$$

Standard deviation

$$\sigma = \sqrt{12}$$

$$\sigma \approx 3.464$$

Interpretation

The first click occurs after **4 visitors on average**, but the waiting time has considerable variability, with a standard deviation of approximately **3.46 visitors**.

Problem 11: First Correct Prediction

A classification model has a probability of **70%** of correctly predicting an observation.

Find the probability that the **first correct prediction occurs on the 2nd observation**.

Solution

Given:

$$\begin{aligned} p &= 0.70 \\ q &= 0.30 \end{aligned}$$

We need:

$$P(X = 2)$$

Therefore:

$$P(X = 2) = (0.3)^1(0.7)$$

$$P(X = 2) = 0.21$$

Answer

$$21\%$$

Interpretation

There is a **21% probability that the first correct prediction occurs exactly on the second observation.**

Problem 12: First Model Error

Suppose a machine-learning model has:

$$p = 0.10$$

probability of making an error on each independent prediction.

Find the probability that the **first error occurs after the 5th prediction.**

Important Difference

"First error occurs after the 5th prediction" means:

$$X > 5$$

Therefore, the first 5 predictions must all be correct.

Probability of a correct prediction:

$$q = 1 - 0.10 = 0.90$$

Thus:

$$P(X > 5) = (0.9)^5$$

$$P(X > 5) = 0.59049$$

Answer

59.05%

Data Science Interpretation

There is approximately a **59.05% probability that the model will make no error in its first five predictions.**

Problem 13: Comparing Two Models

Two classification models have different error rates.

Model A

$$p_A = 0.10$$

Model B

$$p_B = 0.05$$

Here, p represents the probability of an error.

Which model has the **larger expected number of predictions before the first error**?

Solution

For a geometric distribution:

$$E(X) = \frac{1}{p}$$

Model A

$$E(X_A) = \frac{1}{0.10}$$

$$E(X_A) = 10$$

Model B

$$E(X_B) = \frac{1}{0.05}$$

$$E(X_B) = 20$$

Answer

Model B

Model B is expected to make approximately **20 predictions before the first error**, compared with 10 for Model A.

Data Science Interpretation

A lower error probability produces a **longer expected waiting time until the first error**.

Problem 14: Customer Acquisition

A marketing campaign has a probability of:

$$p = 0.05$$

that a contacted person becomes a customer.

Find the probability that the company needs to contact **more than 10 people before getting its first customer**.

Solution

We need:

$$P(X > 10)$$

For geometric distribution:

$$P(X > k) = (1 - p)^k$$

Therefore:

$$\begin{aligned} P(X > 10) &= (1 - 0.05)^{10} \\ P(X > 10) &= (0.95)^{10} \\ &\approx 0.5987 \end{aligned}$$

Answer

$$P(X > 10) \approx 59.87\%$$

Interpretation

There is approximately a **59.87% probability that more than 10 people will need to be contacted before acquiring the first customer**.

Problem 15: Practical Data Science Problem — First Fraud Detection

A fraud-detection system processes transactions sequentially.

The probability that any transaction is fraudulent is:

$$p = 0.01$$

Find:

1. The expected transaction number at which the first fraud occurs.
 2. The variance.
 3. The probability that the first fraud occurs exactly on transaction 20.
 4. The probability that at least one fraud occurs within the first 20 transactions.
-

Solution

Given:

$$p = 0.01$$
$$q = 0.99$$

1. Expected transaction number

$$E(X) = \frac{1}{p}$$
$$E(X) = \frac{1}{0.01}$$

$$E(X) = 100$$

So the first fraudulent transaction occurs at transaction **100 on average**.

2. Variance

$$Var(X) = \frac{1-p}{p^2}$$
$$Var(X) = \frac{0.99}{(0.01)^2}$$
$$Var(X) = \frac{0.99}{0.0001}$$

$$Var(X) = 9900$$

3. First fraud exactly on transaction 20

$$P(X = 20) = q^{19}p$$
$$P(X = 20) = (0.99)^{19}(0.01)$$

$$P(X = 20) \approx 0.00826$$

Therefore:

$$P(X = 20) \approx 0.826\%$$

4. At least one fraud within first 20 transactions

$$\begin{aligned} P(X \leq 20) &= 1 - P(X > 20) \\ P(X \leq 20) &= 1 - (0.99)^{20} \\ &\approx 1 - 0.8179 \end{aligned}$$

$$P(X \leq 20) \approx 0.1821$$

Therefore:

$$18.21\%$$

Data Science Interpretation

Although the expected waiting time is **100 transactions**, there is still about an **18.21% probability of observing a fraud within the first 20 transactions**.

This is an excellent example of why **expected value should not be interpreted as a guaranteed waiting time**.

★ Important Geometric Distribution Formulas

If:

$$X \sim \text{Geometric}(p)$$

and X represents the trial number of the first success, then:

Probability of first success on trial x

$$P(X = x) = (1 - p)^{x-1}p$$

Probability that first success occurs after trial k

$$P(X > k) = (1 - p)^k$$

Probability of success within first k trials

$$P(X \leq k) = 1 - (1 - p)^k$$

Mean

$$E(X) = \frac{1}{p}$$

Variance

$$Var(X) = \frac{1 - p}{p^2}$$

Standard deviation

$$\sigma = \frac{\sqrt{1 - p}}{p}$$



Bernoulli → Binomial → Geometric

This is a very important connection for teaching Data Science.

Bernoulli

One binary trial

Example:

Does this customer purchase?

$$X \sim \text{Bernoulli}(p)$$

Binomial

Number of successes in a fixed number of trials

Example:

How many customers purchase among 100 customers?

$$X \sim \text{Binomial}(n, p)$$

Geometric

Number of trials until the first success

Example:

How many customers do we need to observe until the first purchase?

$$X \sim \text{Geometric}(p)$$

So remember:

Bernoulli → One trial

Binomial → Fixed number of trials, count successes

Geometric → Wait until first success



Geometric Distribution in Data Science

Data Science Application	Success	What X Represents
E-commerce	Purchase	Visitor number of first purchase
Marketing	Conversion	Person number of first conversion
Fraud Detection	Fraud	Transaction number of first fraud
ML Classification	Correct prediction	Prediction number of first correct prediction
ML Error Analysis	Error	Prediction number of first error
Customer Churn	Churn	Customer number of first churn
Recommendation Systems	Successful recommendation	Attempt number of first success
Spam Detection	Spam	Email number of first spam

One-line intuition

Binomial asks: "How many successes did we get?"

Geometric asks: "How long do we wait until the first success?"